

Introduction to R, RStudio, & R packages

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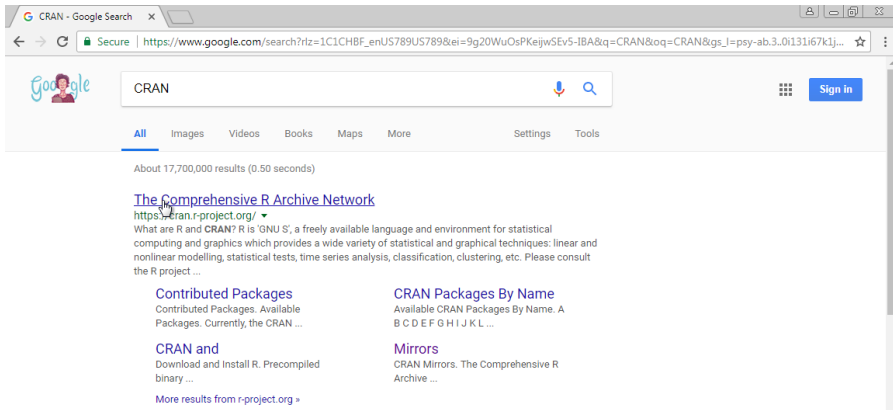
Installing R and RStudio

R is a language for statistical computing and graphics. **RStudio** is an interactive desktop environment (IDE), but it is not R, nor does it include R when you download and install it. Therefore, to use RStudio, we first need to install R.



Installing R (Windows and Mac)

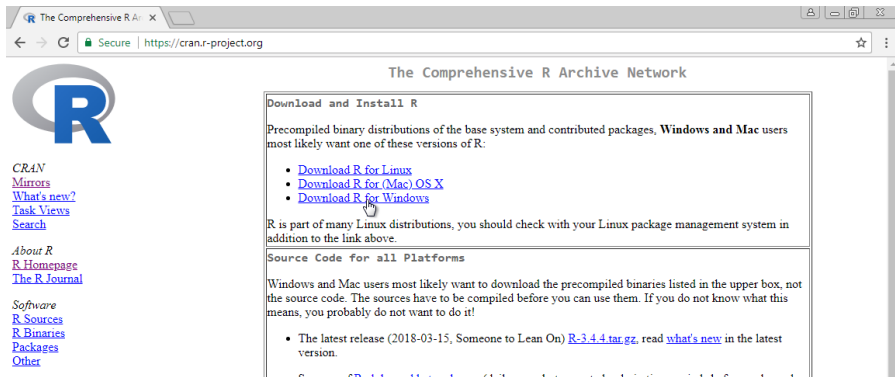
Download R from the Comprehensive R Archive Network (CRAN):
<https://cran.r-project.org/>. Search for “CRAN” on your browser:



The screenshot shows a Google search interface with the query "CRAN". The search results page displays "About 17,700,000 results (0.50 seconds)". The top result is "The Comprehensive R Archive Network" with the URL <https://cran.r-project.org/>. Below the title, a brief description states: "What are R and CRAN? R is 'GNU S', a freely available language and environment for statistical computing and graphics which provides a wide variety of statistical and graphical techniques; linear and nonlinear modelling, statistical tests, time series analysis, classification, clustering, etc. Please consult the R project ...". To the right of the description are two links: "Contributed Packages" and "CRAN Packages By Name". Below the description are two more links: "CRAN and" and "Mirrors". At the bottom, there is a link "More results from r-project.org »".

Installing (Windows and Mac)

Once on the CRAN page, select the version for your operating system: Linux, Mac OS X, or Windows. Here we show screenshots for Windows, but the process is similar for the other platforms. When they differ, we will also show screenshots for Mac OS X.



The screenshot shows the CRAN (Comprehensive R Archive Network) website. The browser address bar shows <https://cran.r-project.org>. The page title is "The Comprehensive R Archive Network". The main content area is titled "Download and Install R". It states: "Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux](#)
- [Download R for \(Mac\) OS X](#)
- [Download R for Windows](#)

Below this, it says: "R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above."

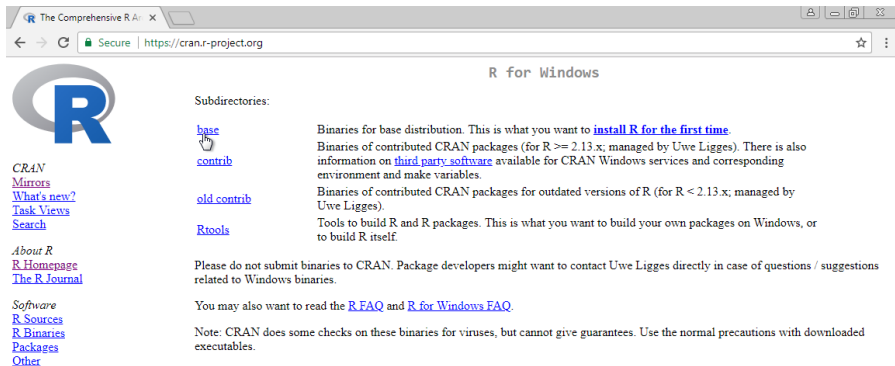
There is a section titled "Source Code for all Platforms". It states: "Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to do it!"

Below this, it lists: "The latest release (2018-03-15, Someone to Lean On) [R-3.4.4 tar.gz](#), read [what's new](#) in the latest version."

On the left side of the page, there is a sidebar with links: "CRAN", "Mirrors", "What's new?", "Task Views", "Search", "About R", "R Homepage", "The R Journal", "Software", "R Sources", "R Binaries", "Packages", "Other".

Installing R (Windows)

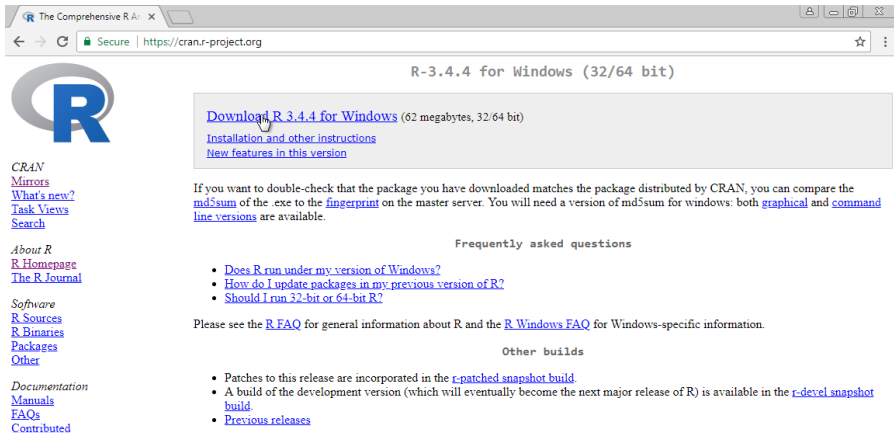
Once at the CRAN download page, you will have several choices. You want to install the *base* subdirectory. This installs the basic packages you need to get started. We will later learn how to install other needed packages from within R.



The screenshot shows a web browser window with the address bar displaying "https://cran.r-project.org". The page title is "R for Windows". On the left side, there is a large blue "R" logo and a list of links: "CRAN", "Mirrors", "What's new?", "Task Views", "Search", "About R", "R Homepage", "The R Journal", "Software", "R Sources", "R Binaries", "Packages", and "Other". The main content area is titled "Subdirectories:" and lists four options: "base", "contrib", "old contrib", and "Rtools". Each option has a brief description. The "base" option is highlighted with a mouse cursor. The "base" description states: "Binaries for base distribution. This is what you want to [install R for the first time](#)." The "contrib" description states: "Binaries of contributed CRAN packages (for R >= 2.13.x; managed by Uwe Ligges). There is also information on [third party software](#) available for CRAN Windows services and corresponding environment and make variables." The "old contrib" description states: "Binaries of contributed CRAN packages for outdated versions of R (for R < 2.13.x; managed by Uwe Ligges)." The "Rtools" description states: "Tools to build R and R packages. This is what you want to build your own packages on Windows, or to build R itself." At the bottom, there is a note: "Please do not submit binaries to CRAN. Package developers might want to contact Uwe Ligges directly in case of questions / suggestions related to Windows binaries." and another note: "You may also want to read the [R FAQ](#) and [R for Windows FAQ](#)." A final note at the bottom states: "Note: CRAN does some checks on these binaries for viruses, but cannot give guarantees. Use the normal precautions with downloaded executables."

Installing R (Windows)

Click on the link for the latest version to start the download.



The screenshot shows the CRAN website for R-3.4.4 for Windows (32/64 bit). The page layout includes a header with the CRAN logo and navigation links on the left. The main content area features a large download link for R 3.4.4 for Windows (62 megabytes, 32/64 bit), which is highlighted with a mouse cursor. Below this, there are links for installation instructions and new features. A section for frequently asked questions is also visible, along with a note about double-checking the package and a link to the R FAQ. The footer contains a note to webmasters about a stable link that will redirect to the current Windows binary release.

CRAN

- [Mirrors](#)
- [What's new?](#)
- [Task Views](#)
- [Search](#)

About R

- [R Homepage](#)
- [The R Journal](#)

Software

- [R Sources](#)
- [R Binaries](#)
- [Packages](#)
- [Other](#)

Documentation

- [Manuals](#)
- [FAQs](#)
- [Contributed](#)

R-3.4.4 for Windows (32/64 bit)

[Download R 3.4.4 for Windows](#) (62 megabytes, 32/64 bit)

[Installation and other instructions](#)

[New features in this version](#)

If you want to double-check that the package you have downloaded matches the package distributed by CRAN, you can compare the [md5sum](#) of the .exe to the [fingerprint](#) on the master server. You will need a version of md5sum for windows: both [graphical](#) and [command line versions](#) are available.

Frequently asked questions

- [Does R run under my version of Windows?](#)
- [How do I update packages in my previous version of R?](#)
- [Should I run 32-bit or 64-bit R?](#)

Please see the [R FAQ](#) for general information about R and the [R Windows FAQ](#) for Windows-specific information.

Other builds

- Patches to this release are incorporated in the [r-patched snapshot build](#).
- A build of the development version (which will eventually become the next major release of R) is available in the [r-devel snapshot build](#).
- [Previous releases](#)

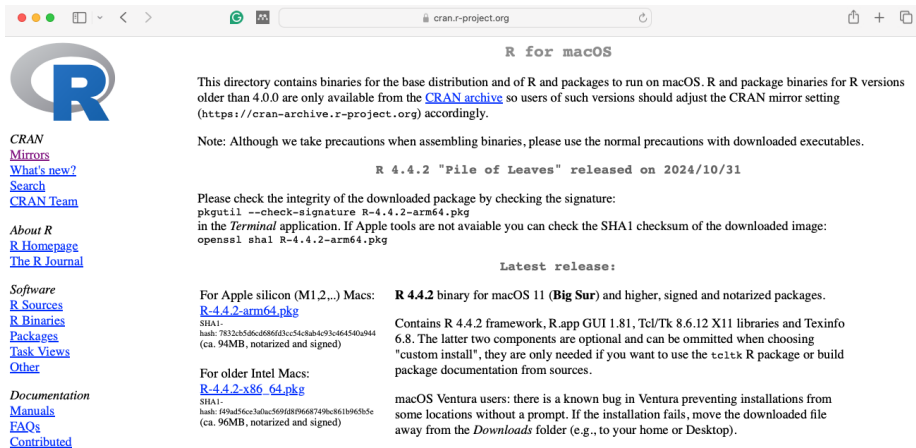
Note to webmasters: A stable link which will redirect to the current Windows binary release is

Installing R (Windows)

Once the installer file downloads, you can click on that tab to start the installation process. Some browsers may display this differently, so you will have to find where they store downloaded files and click on them to get the process started.

Installing R (Mac)

If using Safari on a Mac, things may look a little different:



The screenshot shows the Safari browser window displaying the R for macOS download page. The address bar shows the URL `cran.r-project.org`. The page content includes the R logo, navigation links, and detailed instructions for downloading and verifying the R 4.4.2 binary for macOS 11 (Big Sur) and higher.

R for macOS

This directory contains binaries for the base distribution and of R and packages to run on macOS. R and package binaries for R versions older than 4.0.0 are only available from the [CRAN archive](https://cran-archive.r-project.org) so users of such versions should adjust the CRAN mirror setting (<https://cran-archive.r-project.org>) accordingly.

Note: Although we take precautions when assembling binaries, please use the normal precautions with downloaded executables.

R 4.4.2 "File of Leaves" released on 2024/10/31

Please check the integrity of the downloaded package by checking the signature:

```
pkgutil --check-signature R-4.4.2-arm64.pkg
```

in the *Terminal* application. If Apple tools are not available you can check the SHA1 checksum of the downloaded image:

```
openssl sha1 R-4.4.2-arm64.pkg
```

Latest release:

For Apple silicon (M1,2,...) Macs:

[R-4.4.2-arm64.pkg](#)
SHA1-
hash: 7832cb5d6cd686fd3cc54c8ab4c93c464540a944
(ca. 94MB, notarized and signed)

For older Intel Macs:

[R-4.4.2-x86_64.pkg](#)
SHA1-
hash: f49ad56cc3a0ac569fd8f9668749ec861b965b5e
(ca. 96MB, notarized and signed)

R 4.4.2 binary for macOS 11 (Big Sur) and higher, signed and notarized packages.

Contains R 4.4.2 framework, R.app GUI 1.81, Tcl/Tk 8.6.12 X11 libraries and Texinfo 6.8. The latter two components are optional and can be omitted when choosing "custom install", they are only needed if you want to use the `teitex` R package or build package documentation from sources.

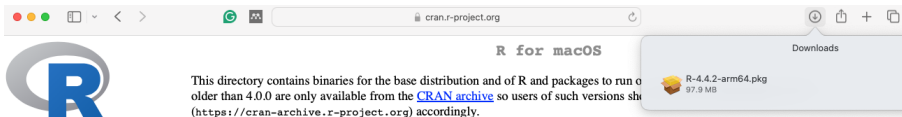
macOS Ventura users: there is a known bug in Ventura preventing installations from some locations without a prompt. If the installation fails, move the downloaded file away from the *Downloads* folder (e.g., to your home or Desktop).

Navigation Links:

- CRAN
- [Mirrors](#)
- [What's new?](#)
- [Search](#)
- [CRAN Team](#)
- About R
- [R Homepage](#)
- [The R Journal](#)
- Software
 - [R Sources](#)
 - [R Binaries](#)
 - [Packages](#)
 - [Task Views](#)
 - [Other](#)
- Documentation
 - [Manuals](#)
 - [FAQs](#)
 - [Contributed](#)

Installing R (Mac)

If using Safari on a Mac, you can access the download through the download button.



CRAN

[Mirrors](#)

[What's new?](#)

[Search](#)

[CRAN Team](#)

About R

[R Homepage](#)

[The R Journal](#)

Software

[R Sources](#)

[R Binaries](#)

[Packages](#)

[Task Views](#)

[Other](#)

Documentation

[Manuals](#)

[FAQs](#)

[Contributed](#)

This directory contains binaries for the base distribution and of R and packages to run on older than 4.0.0 are only available from the [CRAN archive](https://cran-archive.r-project.org) so users of such versions should use the [CRAN archive](https://cran-archive.r-project.org) (https://cran-archive.r-project.org) accordingly.

Note: Although we take precautions when assembling binaries, please use the normal precautions with downloaded executables.

R 4.4.2 "File of Leaves" released on 2024/10/31

Please check the integrity of the downloaded package by checking the signature:

```
pkgutil --check-signature R-4.4.2-arm64.pkg
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in the *Terminal* application. If Apple tools are not available you can check the SHA1 checksum of the downloaded image:

```
openssl sha1 R-4.4.2-arm64.pkg
```

Latest release:

For Apple silicon (M1,2,...) Macs:

[R-4.4.2-arm64.pkg](#)

SHA1-

hash: 7832cb5d6cd686fd3cc54c8ab4c93c464540a944
(ca. 94MB, notarized and signed)

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[R-4.4.2-x86_64.pkg](#)

SHA1-

hash: f49ad56ce3a0ac569fd8f9668749bc861b965b5c
(ca. 96MB, notarized and signed)

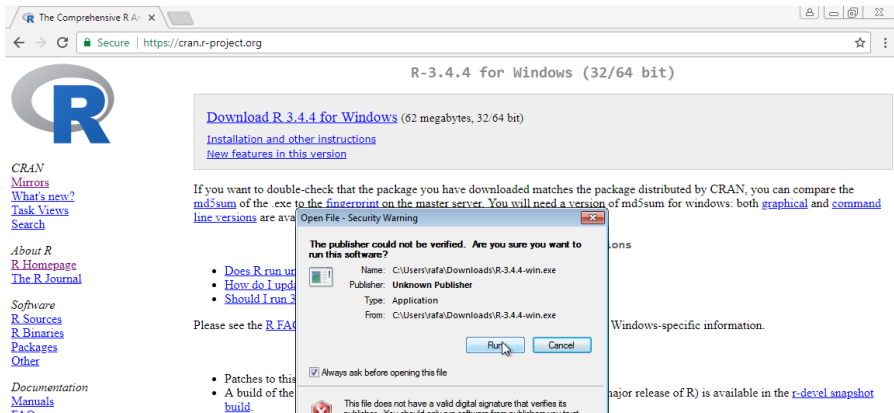
R 4.4.2 binary for macOS 11 (**Big Sur**) and higher, signed and notarized packages.

Contains R 4.4.2 framework, R.app GUI 1.81, Tcl/Tk 8.6.12 X11 libraries and Texinfo 6.8. The latter two components are optional and can be omitted when choosing "custom install", they are only needed if you want to use the `tcltk` R package or build package documentation from sources.

macOS Ventura users: there is a known bug in Ventura preventing installations from some locations without a prompt. If the installation fails, move the downloaded file away from the *Downloads* folder (e.g., to your home or Desktop).

Installing R (Windows)

You can now click through different choices to finish the installation. We recommend you select all the default choices.



The screenshot shows the CRAN website for R-3.4.4 for Windows (32/64 bit). The page includes links for downloading the installer, installation instructions, and new features. A security warning dialog box is overlaid on the page, asking if the user wants to run the software from an unknown publisher.

R-3.4.4 for Windows (32/64 bit)

[Download R 3.4.4 for Windows](#) (62 megabytes, 32/64 bit)

[Installation and other instructions](#)

[New features in this version](#)

If you want to double-check that the package you have downloaded matches the package distributed by CRAN, you can compare the [md5sum](#) of the .exe to the [fingerprint on the master server](#). You will need a version of md5sum for windows: both [graphical](#) and [command line versions](#) are available.

Open File - Security Warning

The publisher could not be verified. Are you sure you want to run this software?

Name: C:\Users\rafa\Downloads\R-3.4.4-win.exe
 Publisher: Unknown Publisher
 Type: Application
 From: C:\Users\rafa\Downloads\R-3.4.4-win.exe

☒ Always ask before opening this file

This file does not have a valid digital signature that verifies its publisher.

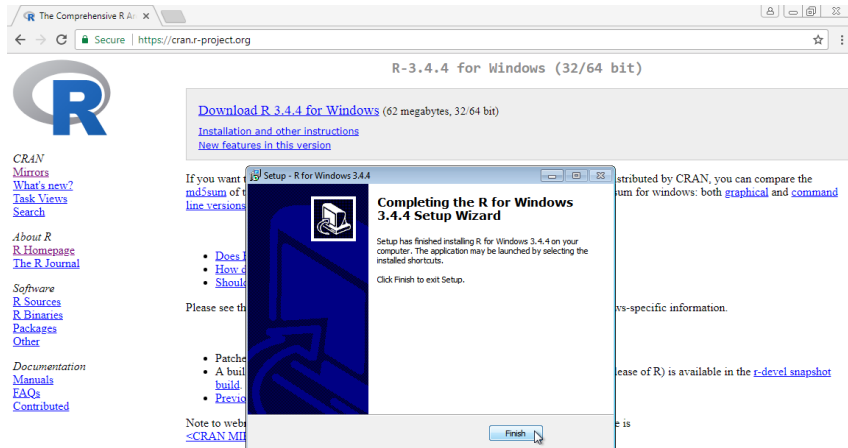
Please see the [R FAQ](#) for more information.

Windows-specific information.

major release of R) is available in the [r-devel snapshot](#)

Installing R (Windows)

Continue to select all the defaults until you have completed the setup:

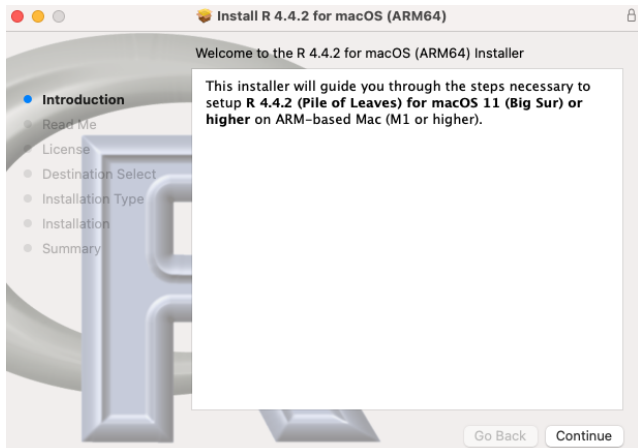


The screenshot shows the CRAN website at <https://cran.r-project.org> with the title "R-3.4.4 for Windows (32/64 bit)". The website includes links for downloading R 3.4.4 for Windows (62 megabytes, 32/64 bit), installation and other instructions, and new features in this version. On the left, there are links for CRAN Mirrors, What's new?, Task Views, Search, About R, R Homepage, The R Journal, Software, R Sources, R Binaries, Packages, Other, Documentation, Manuals, FAQs, and Contributed.

Overlaid on the website is the "Setup - R for Windows 3.4.4" window. The window title is "Completing the R for Windows 3.4.4 Setup Wizard". The main text says: "Setup has finished installing R for Windows 3.4.4 on your computer. The application may be launched by selecting the installed shortcuts. Click Finish to exit Setup." There is a "Finish" button at the bottom right of the window.

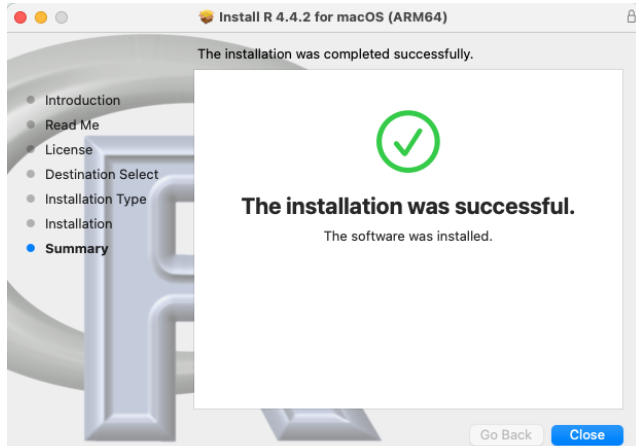
Installing R (Mac)

On the Mac it looks different, but you are again just accepting the defaults:



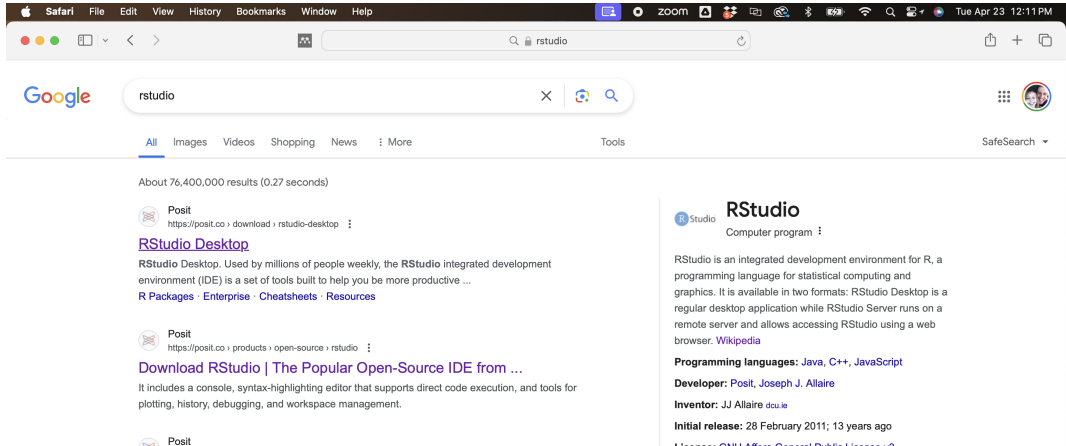
Installing R (Mac)

Congratulations! You have installed R.



Installing RStudio (Windows and Mac)

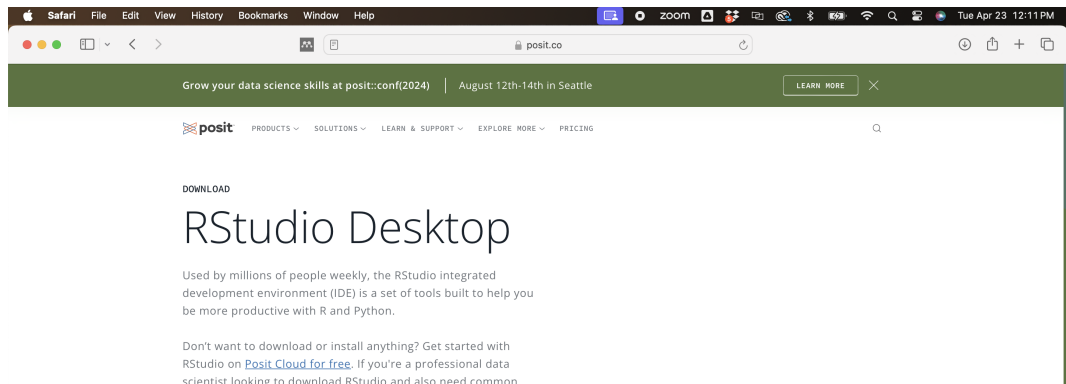
To install RStudio, start by searching for “RStudio” on your browser:



The screenshot shows a Safari browser window with the address bar set to 'rstudio'. The Google search bar contains the text 'rstudio'. The search results page displays 'About 76,400,000 results (0.27 seconds)'. The top result is from Posit, titled 'RStudio Desktop', with a description: 'RStudio Desktop. Used by millions of people weekly, the RStudio integrated development environment (IDE) is a set of tools built to help you be more productive ...'. Below this, there are links for 'R Packages', 'Enterprise', 'Cheatsheets', and 'Resources'. The right sidebar shows a detailed view of the 'RStudio' product page, including its description as an integrated development environment for R, its programming languages (Java, C++, JavaScript), developer (Posit, Joseph J. Allaire), and release date (February 2011).

Installing RStudio (Mac)

You should find the Posit/RStudio website as shown above. Once there, click on “Download RStudio Desktop for Mac OS 12+” below the 2: *Install RStudio* header. We first focus on Mac installation.



Grow your data science skills at [posit::conf\(2024\)](#) | August 12th-14th in Seattle [LEARN MORE](#)

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DOWNLOAD

RStudio Desktop

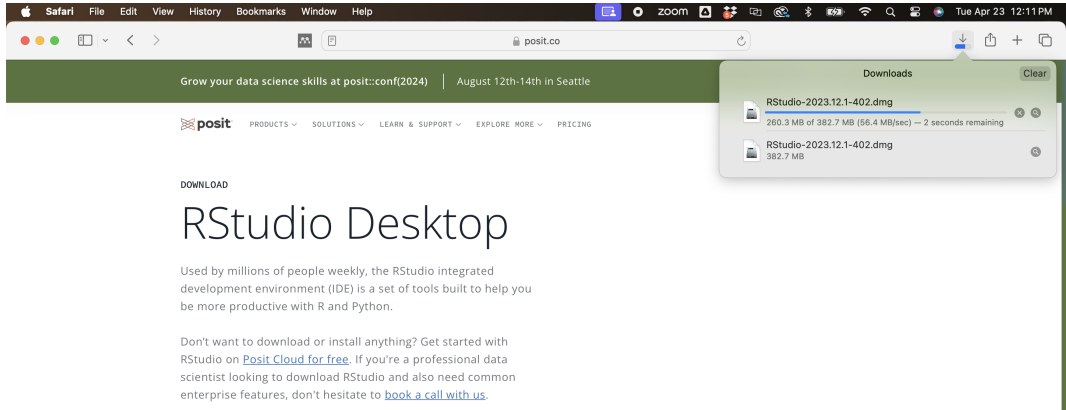
Used by millions of people weekly, the RStudio integrated development environment (IDE) is a set of tools built to help you be more productive with R and Python.

Don't want to download or install anything? Get started with RStudio on [Posit Cloud for free](#). If you're a professional data scientist looking to download RStudio and also need common

Center for Data Science

Installing RStudio (Mac)

The download process should start. For Safari on Mac, you should see the progress in the bar at the top right. Double click on the download when its complete.

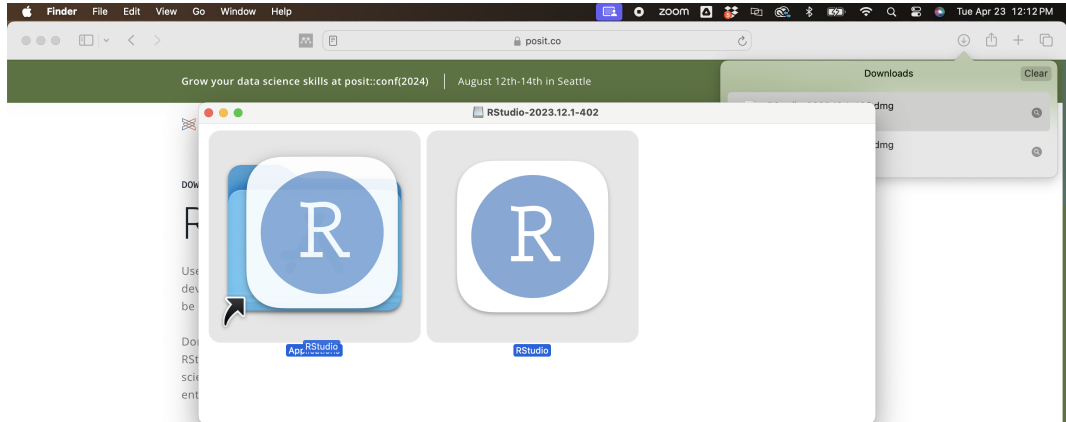


The screenshot shows the Safari browser window with the URL `posit.co`. The page content includes the Posit logo, navigation links (PRODUCTS, SOLUTIONS, LEARN & SUPPORT, EXPLORE MORE, PRICING), and a section titled "DOWNLOAD RStudio Desktop". The text describes RStudio as an integrated development environment (IDE) used by millions of people weekly. It also provides links for downloading RStudio on Posit Cloud for free and booking a call with us.

On the right side of the browser window, the Downloads bar is visible, showing the progress of the download for `RStudio-2023.12.1-402.dmg`. The progress bar indicates that 260.3 MB of 382.7 MB has been downloaded, with a download speed of 56.4 MB/sec and 2 seconds remaining.

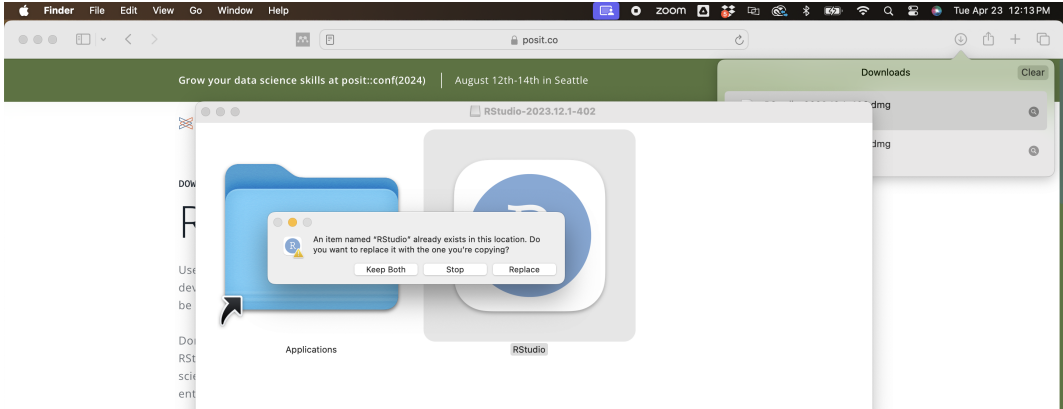
Installing RStudio (Mac)

Now drag and drop the RStudio icon over the Applications icon:



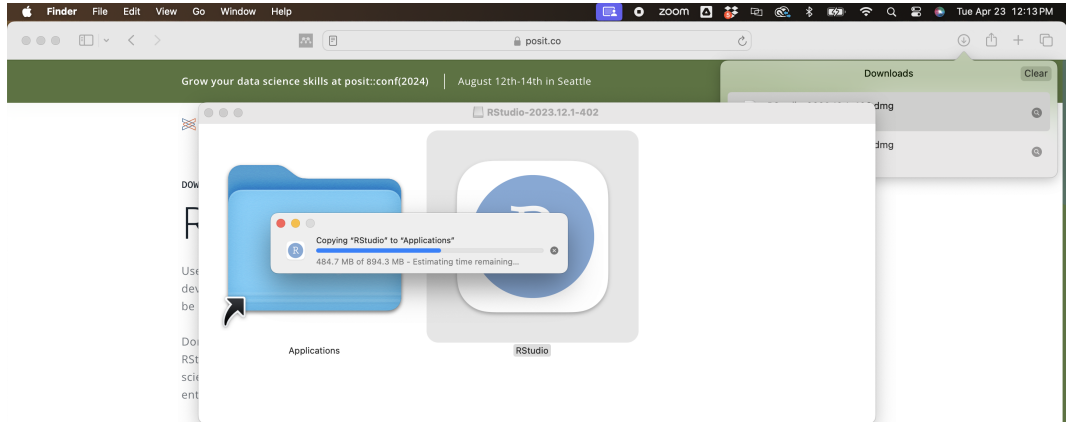
Installing RStudio (Mac)

If you have a previous version of RStudio installed, it may ask if you want to overwrite the old version. We usually want to replace the old with the new, so click “Replace”.



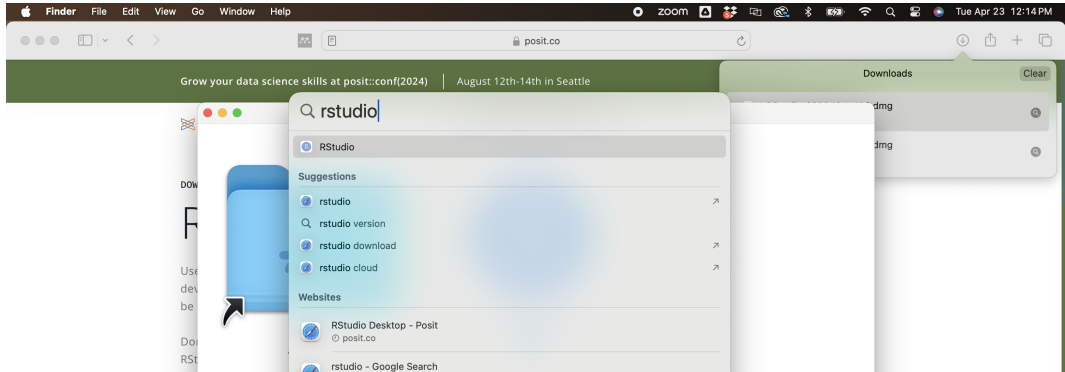
Installing RStudio (Mac)

The system should copy RStudio to the Applications folder:



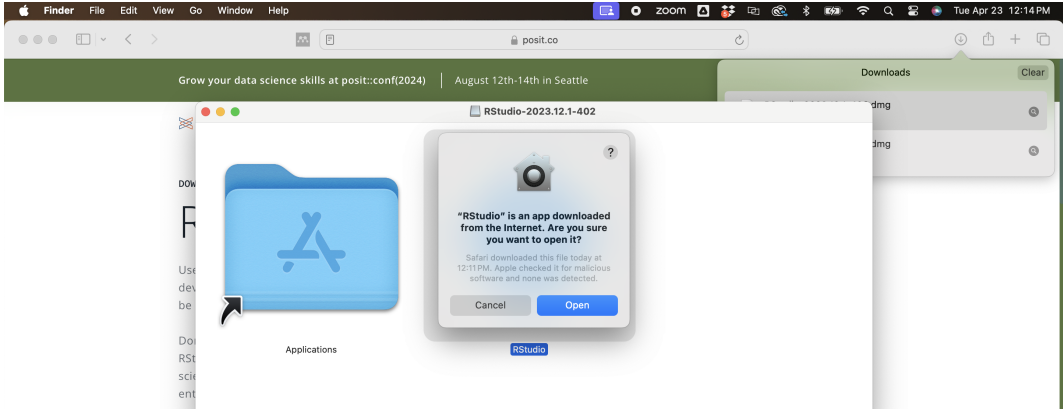
Running RStudio (Mac)

Congratulations! You have installed RStudio. On the Mac, it will be in the Applications folder. Or you can click on the *Spotlight Search* (magnifying glass in top right corner) and type `rstudio` into that search bar, then hit enter.



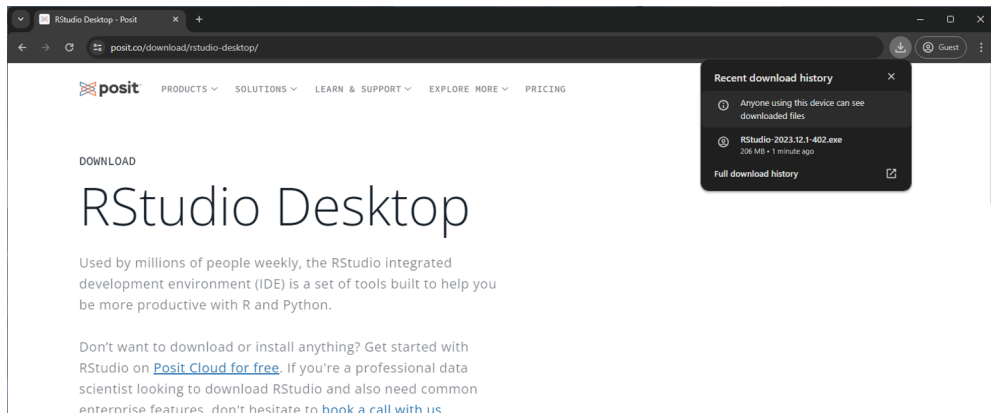
Installing RStudio (Mac)

You may get a warning that RStudio is an application downloaded from the internet. Go ahead and click “Open”.



Installing RStudio (Windows)

For Windows, the process is similar. From the RStudio main page, click on “Download RStudio Desktop for Windows” below the *2: Install RStudio* header.



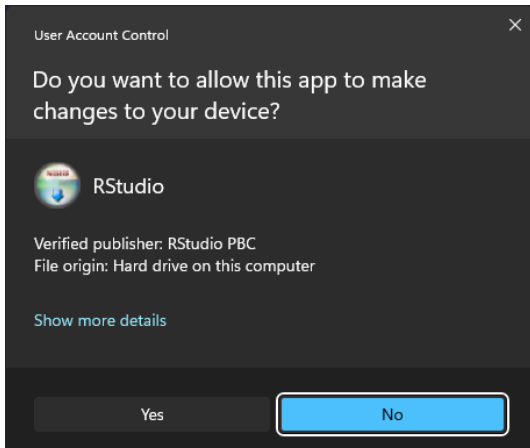
Recent download history

- Anyone using this device can see downloaded files
- RStudio-2023.12.1-402.exe
206 MB • 1 minute ago

Full download history

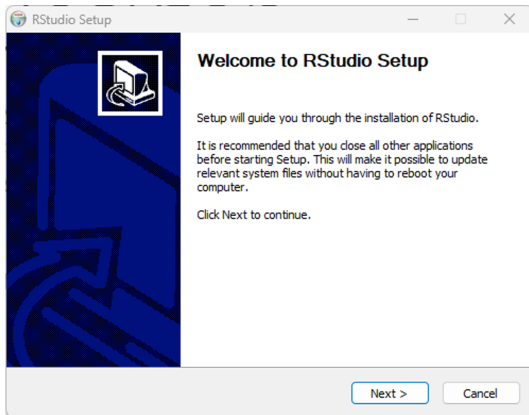
Installing RStudio (Windows)

You may receive a warning, just click “Yes” to continue the install.



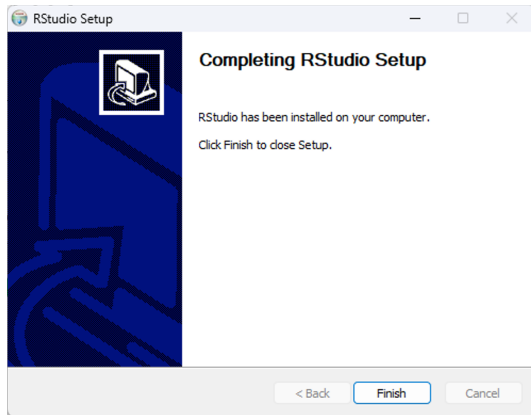
Installing RStudio (Windows)

Follow the instructions to install RStudio, defaults are fine.



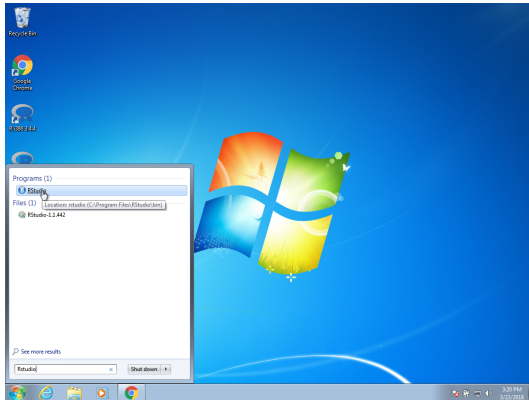
Installing RStudio (Windows)

Congratulations! You are done!



Running RStudio (Windows)

On Windows, you can open RStudio from the *Start* menu:



Getting started: Why R?

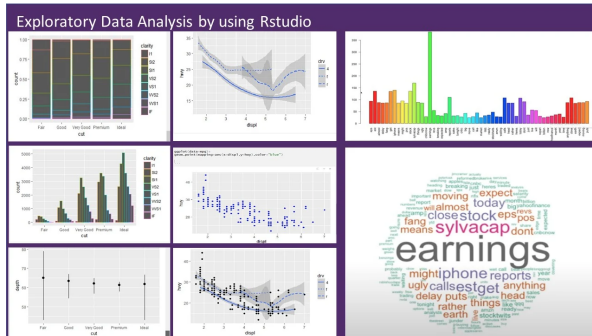
R is not a programming language for software development like C or Java. It was created by statisticians as an environment for data analysis. A history of R is summarized here: [A Brief History of S](#).

The **interactivity** of R (more later), is an indispensable feature in data science because, as you will learn, the ability to quickly explore data is a necessity for success in this field.

Getting started: Why R?

R does not follow conventions of “traditional” programming languages. However, R has unmatched utility when it comes to data analysis and data visualization.

A history of R is summarized here: [A Brief History of S](#).



Getting started: Why R?

Other attractive features of R are:

1. R is free and open source.
2. It runs on all major platforms: Windows, Mac Os, UNIX/Linux.
3. Scripts and data objects can be shared seamlessly across platforms.
4. There is a large, growing, and active community of R users and, as a result, there are numerous resources for learning and asking questions^{1 2 3}.
5. It is easy for others to contribute add-ons which enables developers to share software implementations of new data science methodologies.

¹<https://www.r-project.org/help.html>

²<https://stackoverflow.com/documentation/r/topics>

³<https://stats.stackexchange.com/questions/138/free-resources-for-learning-r>

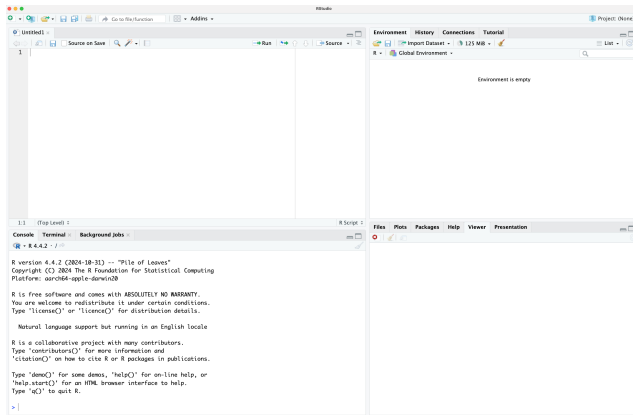
RStudio

RStudio will be our launching pad for data science projects. It provides an editor for our scripts and provides many other useful tools. Here are some of the basics.



RStudio Panes

When you start RStudio, you will see three or four panes:



The R console

Interactive data analysis usually occurs on the *R console*. As a quick example, try using the console to calculate a 15% tip on a meal that cost \$19.71:

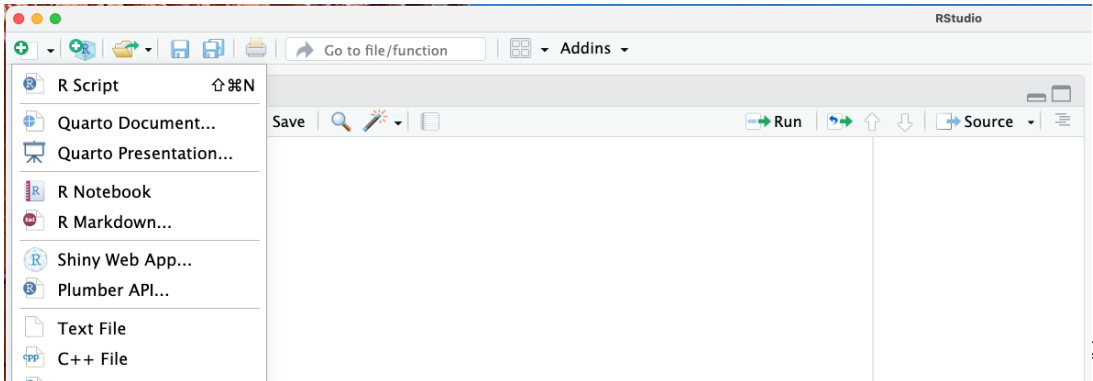
```
0.15 * 19.71
```

```
## [1] 2.9565
```

Pro Tip: Grey boxes in the lecture notes are used to show R code typed into the R console. The symbol `##` is used to denote what the R console outputs.

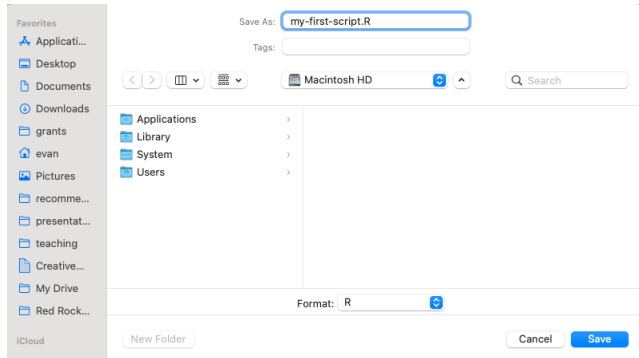
Running commands while editing scripts

Let's start by opening a new script and giving the script a name. We can do this through the editor by saving the current new unnamed script.



Running commands while editing scripts

When saving a script, a good convention is to use a descriptive name, with lower case letters, no spaces, only hyphens to separate words, and then followed by the suffix *.R*. We will call this script *my-first-script.R*.



Running commands while editing scripts

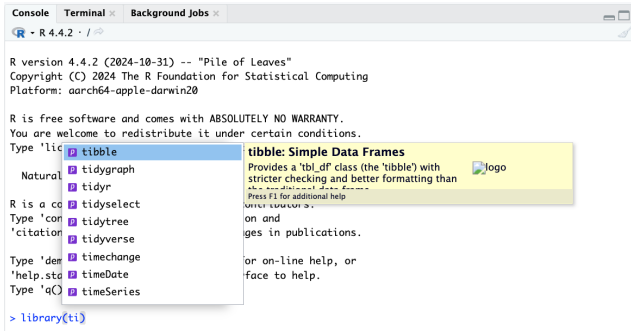
Now we are ready to start running our first script. There are many ways to execute a script in R/RStudio:

1. **Run Line by Line:** Place the cursor on a line and press **Ctrl+Enter** (Windows/Linux) or **Cmd+Enter** (Mac).
2. **Select and Run:** Highlight code and press **Ctrl+Enter** (Windows/Linux) or **Cmd+Enter** (Mac).
3. **Run Entire Script:** Click **Source** or press **Ctrl+Shift+Enter** (Windows/Linux) or **Cmd+Shift+Enter** (Mac).
4. **Use `source()`:** Execute the script from the console with:

```
source("path/to/your_script.R")
```

Organizing scripts

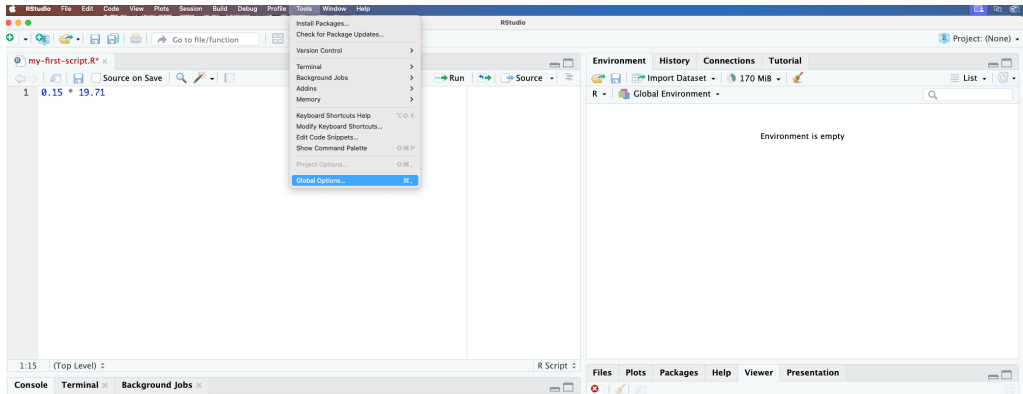
The first lines of code in an R script are dedicated to loading the libraries we will use. One useful feature is RStudio's auto-complete: note what happens when we type `library(ti)`:



Changing global options

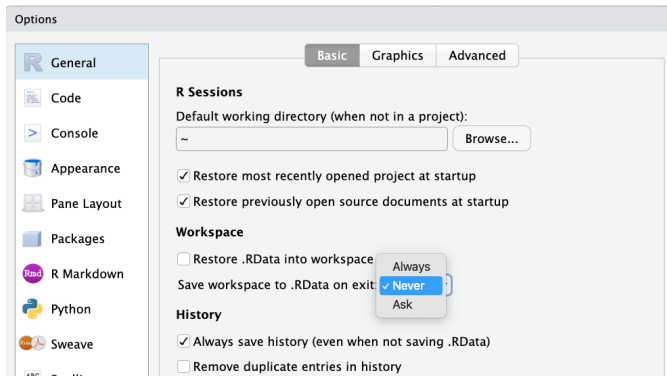
You can change the look and functionality of RStudio quite a bit.

To change global options you click on *Tools* then *Global Options*:



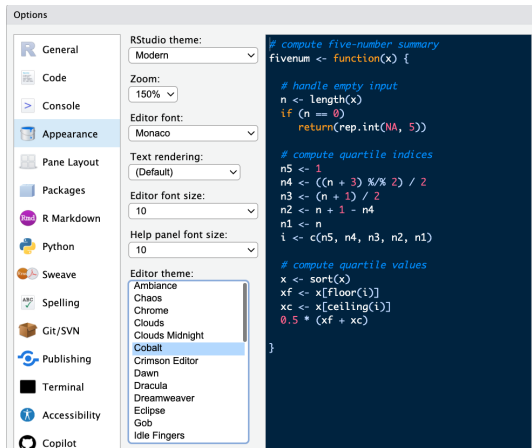
Changing global options

I **highly recommend** to change the *Save workspace to .RData on exit* to *Never* and uncheck the *Restore .RData into workspace at start*. To change these options, make your *General* settings look like this:



Changing global options

Also, click on *Appearance* then try the *Cobalt* option for the *Editor theme*:



Installing R packages

The functionality provided by a fresh install of R is only a small fraction of what is possible. In fact, we refer to what you get after your first install as **base R**. The extra functionality comes from add-ons available from developers.

There are currently hundreds of these available from CRAN and many others shared via other repositories such as GitHub. However, because not everybody needs all available functionality, R instead makes different components available via **packages**.

Installing R packages

R makes it very easy to install packages from within R. For example, to install the **tidyverse** package, which we use to wrangle data in later sessions:

```
install.packages("tidyverse")
```

We can install more than one package at once by feeding a character vector to this function:

```
install.packages(c("tidyverse", "ggplot2"))
```

Installing R packages (Windows)

If you are using Windows, you may get a warning that says:

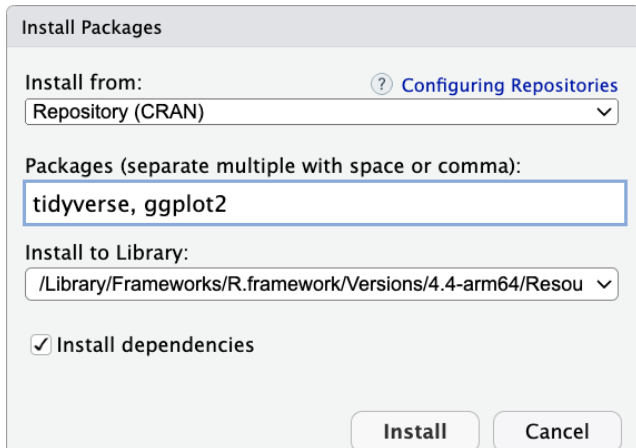
`WARNING: Rtools is required to build R packages but is not currently installed.`

Rtools is a toolchain bundle used for building R packages from source (those that need compilation of C/C++ or Fortran code) and to build R itself from source. We won't cover this here, but for a detailed step-by-step instruction tutorial can be found by clicking the following link:

https://jtleek.com/modules/01_DataScientistToolbox/02_10_rtools/

Installing R packages

In RStudio, you can navigate to the **Tools** tab and select install packages this way:



The image shows the 'Install Packages' dialog box in RStudio. It has a title bar 'Install Packages'. Below the title bar, there are three main sections. The first section is 'Install from:', which has a dropdown menu currently showing 'Repository (CRAN)'. To the right of this dropdown is a blue link with a question mark icon that says 'Configuring Repositories'. The second section is 'Packages (separate multiple with space or comma):', which has a text input field containing 'tidyverse, ggplot2'. The third section is 'Install to Library:', which has a dropdown menu showing '/Library/Frameworks/R.framework/Versions/4.4-arm64/Resou'. Below these sections is a checkbox labeled 'Install dependencies' which is checked. At the bottom right of the dialog are two buttons: 'Install' and 'Cancel'.

Install Packages

Install from: [? Configuring Repositories](#)
Repository (CRAN) ▼

Packages (separate multiple with space or comma):
tidyverse, ggplot2

Install to Library:
/Library/Frameworks/R.framework/Versions/4.4-arm64/Resou ▼

☒ Install dependencies

Install Cancel

Installing R packages

Remember packages are installed in R not RStudio. Also note that installing **tidyverse** actually installs several packages. This commonly occurs when a package has **dependencies**, or uses functions from other packages. When you load a package using `library`, you also load its dependencies.

Installing R packages

Once packages are installed, you can load them into R using the `library` function:

```
library(tidyverse)
```

As you go through this tutorial, you will see that we load packages without installing them. This is because once you install a package, it remains installed and only needs to be loaded with `library`. The package remains loaded until we quit the R session. If you try to load a package and get an error, it probably means you need to install it first.

Installing R packages

Pro Tip: It is helpful to keep a list of all the packages you need for your work in a script because if you need to perform a fresh install of R, you can re-install all your packages by simply running a script.

You can see all the packages you have installed using the following function:

```
installed.packages()
```

Session Info

The `sessionInfo()` function reports information about your R session. Its always good to end your analyses and reports with this function for reproducibility.

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.5.1
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/libBLAS.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Denver
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## loaded via a namespace (and not attached):
## [1] compiler_4.5.2    fastmap_1.2.0     cli_3.6.6         tools_4.5.2
## [5] htmltools_0.5.9   otel_0.2.0        rstudioapi_0.19.0 yaml_2.3.12
## [9] rmarkdown_2.31    knitr_1.51        xfun_0.58         digest_0.6.39
```